

Factorial Designs

Two-way anova with interactions

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Learning Objectives

By the end of this lectures (we will probably spend two lectures on this), you should be able to:

- Explain the advantage of factorial designs over one-factor-at-a-time experiments
- Construct the statistical model for a two-factor factorial: $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$
- Design a factorial experiment using CRD or RCBD structure
- Create and interpret interaction plots to visualize how factors combine
- Conduct simple effects analysis when significant interactions are present
- Make appropriate statistical conclusions when interactions are significant vs non-significant

Packages Used

```
library(dplyr) # Data wrangling across examples
library(tidyr) # Reshaping data (herbicide example)
library(ggplot2) # Plots throughout
library(gridExtra) # Side-by-side layout comparison
library(FieldHub) # Factorial CRD/RCBD design generation
library(agricolae) # Herbicide RCBD design
library(car) # Type III ANOVA tables
library(emmeans) # Estimated marginal means and comparisons
library(parameters) # Model summaries (herbicide ANOVA)
library(performance) # Model diagnostics
library(multcomp) # Compact letter display
```

Motivating Question

Research scenario:

You want to study the effects of **irrigation** and **nitrogen fertilizer** on corn yield.

Two approaches:

1. Run **two separate experiments** (one-factor-at-a-time)
2. Run **one factorial experiment** (both factors together)

Which approach is better?

Review: Single-Factor Designs

CRD (Completely Randomized Design):

- One factor (treatment) with multiple levels
- Randomize treatments to experimental units
- ANOVA: Test treatment effect

Example: Nitrogen (0, 50, 100 kg/ha) on wheat yield

RCBD (Randomized Complete Block Design):

- One factor + blocking for spatial/temporal variation
- Complete set of treatments in each block
- ANOVA: Test treatment effect (accounting for blocks)

Example: Same nitrogen study, blocked by field location

Limitation: Both designs test **one factor at a time**. What if you want to test irrigation AND nitrogen simultaneously?

Why Factorial Designs?

One-Factor-At-a-Time (OFAT) Approach

Experiment 1: Test Irrigation

- Low irrigation (L) vs High irrigation (H)
- Keep nitrogen constant at 50 kg/ha
- 2 treatments $\times$ 8 reps = **16 plots**

Result: High irrigation increases yield by 500 kg/ha

Experiment 2: Test Nitrogen

- No nitrogen (0) vs High nitrogen (100 kg/ha)
- Keep irrigation constant at Low
- 2 treatments $\times$ 8 reps = **16 plots**

Result: Nitrogen increases yield by 300 kg/ha

Problem with OFAT:

- **Total plots:** 32
- **Conclusion:** “Add high irrigation (+500 kg/ha) and nitrogen (+300 kg/ha) for +800 kg/ha total”
- **Hidden assumption:** Effects are **independent** (additive)
- **Risk:** What if nitrogen only works with high irrigation? **You’ll never know!**

Factorial Solution: Test All Combinations

2 $\times$ 2 Factorial Design:

- Factor A: **Irrigation** (Low, High)
- Factor B: **Nitrogen** (0, 100 kg/ha)
- **Replication:** 8 reps per combination
- **Total plots:** $4 \times 8 = \mathbf{32}$ plots (same as OFAT!)

Treatment combinations: 4 (all possible pairs)

1. Low Irrigation + No Nitrogen (L0)
2. Low Irrigation + High Nitrogen (L100)
3. High Irrigation + No Nitrogen (H0)
4. High Irrigation + High Nitrogen (H100)

Treatment Structure Table:

Irrigation	N = 0	N = 100
Low	L0	L100
High	H0	H100

Factorial advantage:

- Same sample size (32 plots)

- **More information:** Tests irrigation effect, nitrogen effect, AND their interaction!

Factorial Notation

Factorial designs are described by the number of factors and their levels:

- **2×2 factorial:** 2 factors, each with 2 levels → 4 treatment combinations
- **3×3 factorial:** 2 factors, each with 3 levels → 9 treatment combinations
- **2×4 factorial:** 2 factors (2 levels, 4 levels) → 8 treatment combinations
- **2×2×2 factorial:** 3 factors, each with 2 levels → 8 treatment combinations
- **2^k factorial:** k factors, all with 2 levels → 2^k combinations

Examples:

Design	Factors	Levels	Combinations
2×2	Irrigation (2) × Nitrogen (2)	2, 2	4
3×3	Irrigation (3) × Nitrogen (3)	3, 3	9
2×3	Tillage (2) × Variety (3)	2, 3	6
2×2×2	Tillage × Variety × Fungicide	2, 2, 2	8

Main Effects and Interactions

What is a Main Effect?

Main effect of Factor A:

The **average effect** of Factor A **across all levels of Factor B**.

- Compares marginal means (averaged over other factors)
- Ignores specific combinations
- Answers: “On average, does Factor A matter?”

Example: Main effect of Irrigation

Mean yield under Low Irrigation:

$$\bar{y}_{\text{Low}} = \frac{y_{L0} + y_{L100}}{2}$$

Average of L0 and L100 (across both nitrogen levels)

Mean yield under High Irrigation:

$$\bar{y}_{\text{High}} = \frac{y_{H0} + y_{H100}}{2}$$

Average of H0 and H100

Main effect of Irrigation:

$$\text{Main Effect}_{\text{Irrigation}} = \bar{y}_{\text{High}} - \bar{y}_{\text{Low}}$$

What is an Interaction?

Interaction between Factor A and Factor B:

The effect of Factor A **depends on** the level of Factor B (or vice versa).

- Main effects are **not additive**
- Simple effects (A at each level of B) differ
- Answers: “Do factors work together in a non-additive way?”

Example: Irrigation × Nitrogen Interaction

If no interaction:

- Nitrogen effect at Low Irrigation = +200 kg/ha
- Nitrogen effect at High Irrigation = +200 kg/ha
- “Nitrogen increases yield by 200 kg/ha regardless of irrigation level”

If interaction present:

- Nitrogen effect at Low Irrigation = +200 kg/ha
- Nitrogen effect at High Irrigation = +1000 kg/ha
- “Nitrogen benefit is much larger under high irrigation!”
- **Factors interact** (synergy)

Biological interpretation:

Nitrogen uptake requires adequate water. Under drought (low irrigation), plants can’t utilize added nitrogen. Under high irrigation, nitrogen is fully effective → **synergistic interaction**.

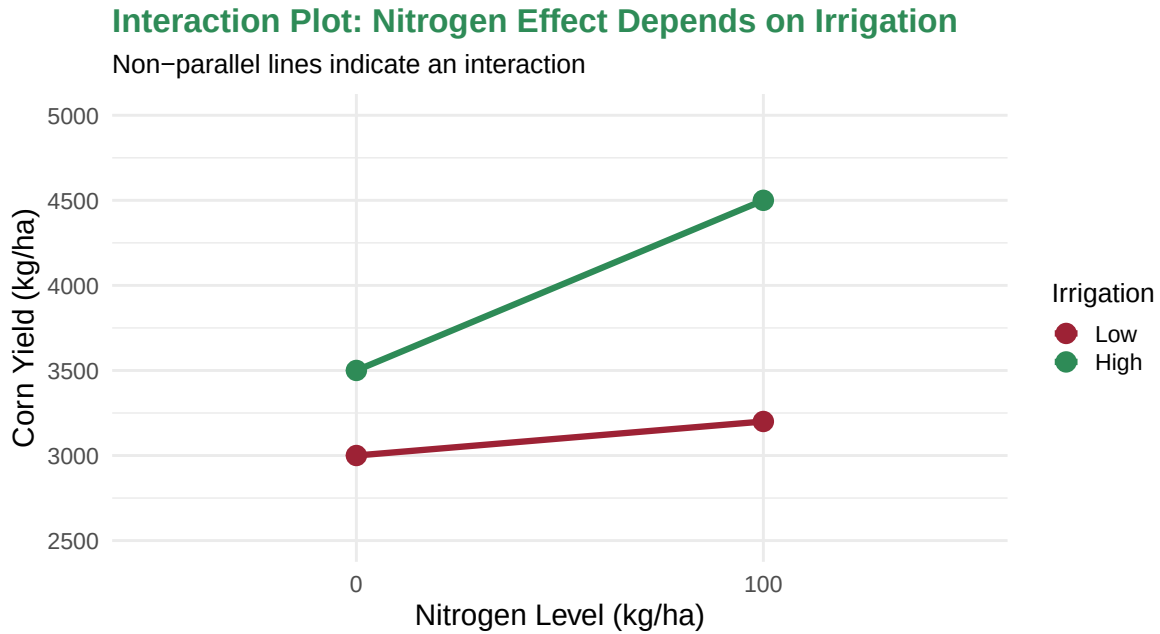


Figure 1: Conceptual interaction plot: Nitrogen effect depends on irrigation

Interaction Plot

Key observation: Lines are **not parallel** → Interaction present!

- Under **low irrigation** (red): Nitrogen increases yield slightly (3000 → 3200)
- Under **high irrigation** (green): Nitrogen increases yield substantially (3500 → 4500)
- **Interpretation:** Nitrogen benefit depends on irrigation level

Statistical Model for Factorial Designs

Two-Factor Factorial Model

Mathematical model for 2×2 factorial:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

Where:

- y_{ijk} = response (e.g., yield) for observation k in treatment i, j
- μ = overall mean (grand mean)
- α_i = main effect of Factor A (e.g., irrigation), $i = 1, 2$
- β_j = main effect of Factor B (e.g., nitrogen), $j = 1, 2$
- $(\alpha\beta)_{ij}$ = interaction effect between Factors A and B
- ε_{ijk} = random error, $\varepsilon_{ijk} \sim N(0, \sigma^2)$

Assumptions

Same assumptions as CRD and RCBD:

1. **Independence:** Observations are independent (proper randomization)
2. **Normality:** Errors ε_{ijk} are normally distributed
3. **Homogeneity of variance:** Equal variance across all treatment combinations (σ^2 constant)
4. **Additivity:** Model is additive (or check for interaction if suspected)

Checking assumptions:

- **Residual plots** (normality, constant variance, outliers)
- **Shapiro-Wilk test** (normality)
- **Levene's test** (homogeneity of variance)

ANOVA for Factorial Design

ANOVA partitions total variation into:

$$SS_{\text{Total}} = SS_A + SS_B + SS_{A \times B} + SS_{\text{Error}}$$

Sources of variation:

1. **Main effect of Factor A** (irrigation)
2. **Main effect of Factor B** (nitrogen)
3. **Interaction A $\times$ B** (irrigation $\times$ nitrogen)
4. **Error** (within-treatment variation)

ANOVA Table Structure:

Source	df	SS	MS	F	p-value
Factor A	$a - 1$	SS_A	MS_A	$\frac{MS_A}{MS_E}$	
Factor B	$b - 1$	SS_B	MS_B	$\frac{MS_B}{MS_E}$	
A×B	$(a - 1)(b - 1)$	$SS_{A \times B}$	$MS_{A \times B}$	$\frac{MS_{A \times B}}{MS_E}$	
Error	$ab(r - 1)$	SS_E	MS_E		
Total	$abr - 1$	SS_{Total}			

Hypotheses Tested

Main effect of Factor A (Irrigation):

$$H_0 : \alpha_1 = \alpha_2 = 0$$

$$H_A : \text{At least one } \alpha_i \neq 0$$

“Is there an average difference between irrigation levels?”

Main effect of Factor B (Nitrogen):

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{At least one } \beta_j \neq 0$$

“Is there an average difference between nitrogen levels?”

Interaction A×B (Irrigation × Nitrogen):

$$H_0 : (\alpha\beta)_{ij} = 0 \text{ for all } i, j$$

$$H_A : \text{At least one } (\alpha\beta)_{ij} \neq 0$$

“Does the effect of nitrogen depend on irrigation level (or vice versa)?”

Designing a Factorial Experiment

Factorial + CRD Structure

Factorial CRD:

- Treatment structure: Factorial (e.g., $2 \times 2 = 4$ combinations)
- Design structure: Completely randomized (no blocks)
- Randomize all treatment combinations to experimental units

When to use:

- Homogeneous experimental environment
- No obvious blocking factor

Example R code with FieldHub:

```
# Generate 2x2 factorial CRD
design_factorial_crd <- FieldHub::CRD(
  t = 4, # 4 treatment combinations (2x2)
  reps = 8, # 8 replicates
  plotNumber = 101,
  seed = 2026,
  locationName = "Research Farm"
)

# View layout
print(design_factorial_crd)
plot(design_factorial_crd)
```

Factorial + RCBD Structure

Factorial RCBD:

- Treatment structure: Factorial (e.g., $2 \times 2 = 4$ combinations)
- Design structure: Randomized complete block
- Each block contains all 4 treatment combinations (complete)

When to use:

- Heterogeneous environment (field gradient, time variability)
- Want to control for blocking factor

Example R code with FieldHub:

```
# Generate 2x2 factorial RCBD
design_factorial_rcbd <- FieldHub::RCBD(
  t = 4, # 4 treatment combinations (2x2)
  reps = 8, # 8 blocks
  plotNumber = 101,
  seed = 2026,
  locationName = "Field Trial"
)

# View layout
print(design_factorial_rcbd)
plot(design_factorial_rcbd)
```

Design Considerations

Key design decisions for factorials:

1. How many factors?

- Start with 2-3 factors (more factors = exponential growth in plots)
- Example: $2 \times 2 \times 2 = 8$ combinations; $3 \times 3 \times 3 = 27$ combinations

2. How many levels per factor?

- 2 levels: Compare two conditions (yes/no, low/high)
- 3+ levels: Dose-response, multiple varieties

3. Replication:

- Need sufficient power: 6-10 reps per combination typical
- Power analysis: Estimate from pilot data or literature

4. Design structure:

- CRD if environment homogeneous
- RCBD if spatial/temporal variation expected

What is an Interaction?

Interaction: Definition

Interaction between Factor A and Factor B:

The effect of Factor A **depends on** the level of Factor B (or equivalently, the effect of Factor B depends on the level of Factor A).

Synonyms:

- Non-additive effect
- Synergistic/antagonistic effect
- Conditional effect
- Moderation

Key insight: When an interaction is present, you **cannot interpret main effects in isolation**—you must consider factor combinations.

No Interaction vs Interaction

Scenario 1: No Interaction (Additive)

Irrigation	N=0	N=100	N effect
Low	3000	3400	+400
High	3500	3900	+400
Irrig effect	+500	+500	

Nitrogen effect is **the same** (+400) at both irrigation levels
Irrigation effect is **the same** (+500) at both nitrogen levels
Effects are **additive**

Scenario 2: Interaction Present

Irrigation	N=0	N=100	N effect
Low	3000	3200	+200
High	3500	4500	+1000
Irrig effect	+500	+1300	

🔥 Caution

Nitrogen effect **differs** (+200 vs +1000)
 Irrigation effect **differs** (+500 vs +1300)
 Effects are **non-additive** → **Interaction!**

Visual Check: Interaction Plots

Interaction plot:

- X-axis: One factor (e.g., Nitrogen)
- Y-axis: Response (e.g., Yield)
- Lines: Levels of the other factor (e.g., Irrigation)

Interpretation rule:

- **Parallel lines** → No interaction (additive effects)
- **Non-parallel lines** → Interaction present (non-additive effects)

Interaction Plot: No Interaction

```
library(ggplot2)
library(dplyr)

# Data: No interaction (additive)
no_interaction <- data.frame(
  Irrigation = rep(c("Low", "High"), each = 2),
  Nitrogen = rep(c("0", "100"), times = 2),
  Yield = c(3000, 3400, 3500, 3900) # N adds 400 everywhere
)

no_interaction$Irrigation <- factor(no_interaction$Irrigation, levels = c("Low", "High"))
no_interaction$Nitrogen <- factor(no_interaction$Nitrogen, levels = c("0", "100"))
```

```

# Interaction plot
ggplot(no_interaction, aes(x = Nitrogen, y = Yield,
                           color = Irrigation, group = Irrigation)) +
  geom_line(linewidth = 2) +
  geom_point(size = 6) +
  labs(
    title = "No Interaction: Parallel Lines",
    subtitle = "Nitrogen effect is the same (+400 kg/ha) under both irrigation levels",
    x = "Nitrogen Level (kg/ha)",
    y = "Corn Yield (kg/ha)",
    color = "Irrigation"
  ) +
  scale_color_manual(values = c("Low" = "#9D2235", "High" = "#2E8B57")) +
  scale_y_continuous(limits = c(2500, 4500), breaks = seq(2500, 4500, 500)) +
  theme_minimal(base_size = 18) +
  theme(
    plot.title = element_text(color = "#2E8B57", face = "bold", size = 20),
    plot.subtitle = element_text(size = 16),
    legend.position = "right",
    legend.title = element_text(size = 16),
    legend.text = element_text(size = 14)
  )

```

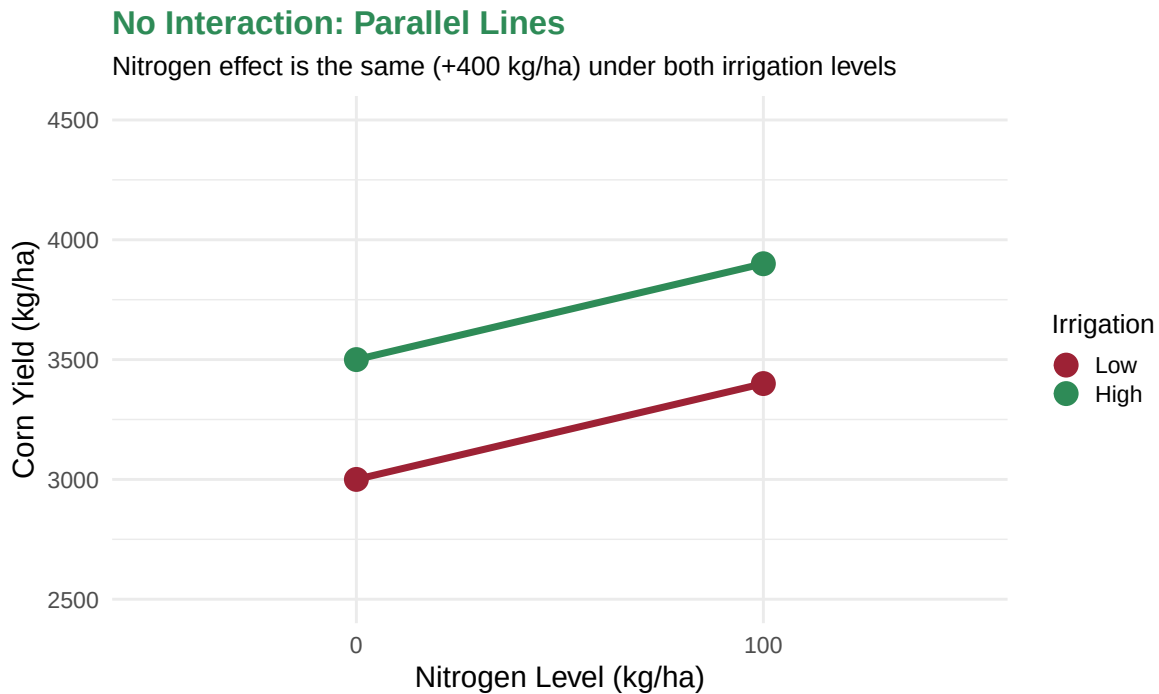


Figure 2: No interaction: Parallel lines indicate additive effects

Parallel lines: Slope is identical for both irrigation levels → **No interaction.** You can interpret main effects directly.

Interaction Plot: Synergistic Interaction

```
# Data: Synergistic interaction
synergistic <- data.frame(
  Irrigation = rep(c("Low", "High"), each = 2),
  Nitrogen = rep(c("0", "100"), times = 2),
  Yield = c(3000, 3200, 3500, 4500) # N effect larger under high irrigation
)

synergistic$Irrigation <- factor(synergistic$Irrigation, levels = c("Low", "High"))
synergistic$Nitrogen <- factor(synergistic$Nitrogen, levels = c("0", "100"))

# Interaction plot
ggplot(synergistic, aes(x = Nitrogen, y = Yield,
  color = Irrigation, group = Irrigation)) +
```

```

geom_line(linewidth = 2) +
geom_point(size = 6) +
labs(
  title = "Synergistic Interaction: Nitrogen Works Better with High Irrigation",
  subtitle = "Non-parallel lines diverge: Nitrogen effect depends on irrigation level",
  x = "Nitrogen Level (kg/ha)",
  y = "Corn Yield (kg/ha)",
  color = "Irrigation"
) +
scale_color_manual(values = c("Low" = "#9D2235", "High" = "#2E8B57")) +
scale_y_continuous(limits = c(2500, 5000), breaks = seq(2500, 5000, 500)) +
theme_minimal(base_size = 18) +
theme(
  plot.title = element_text(color = "#2E8B57", face = "bold", size = 20),
  plot.subtitle = element_text(size = 16),
  legend.position = "right",
  legend.title = element_text(size = 16),
  legend.text = element_text(size = 14)
)

```

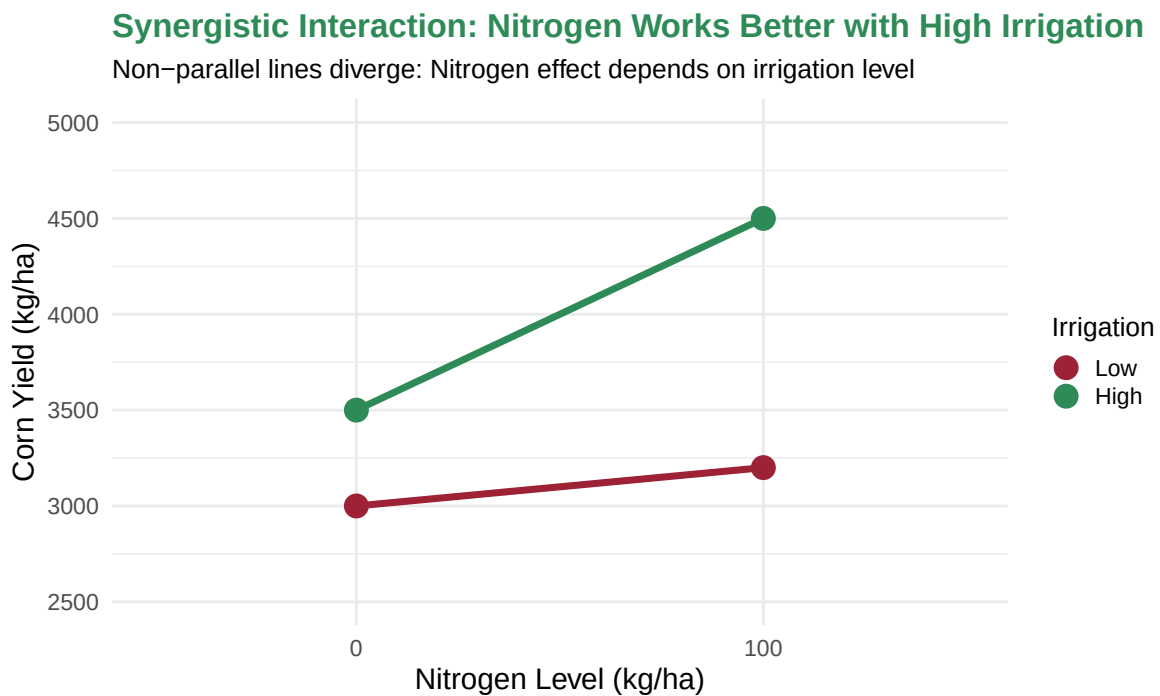


Figure 3: Synergistic interaction: Nitrogen effect amplified under high irrigation

Non-parallel lines (diverging): Green line (high irrigation) has steeper slope than red line (low irrigation) → **Synergistic interaction**. Factors amplify each other's effects.

Types of Interactions

Synergistic Interaction

Synergistic interaction:

The combined effect of two factors is **greater than** the sum of their individual effects.

Visual cue: Non-parallel lines that **diverge** (spread apart)

Example: Fungicide + resistant variety control disease better together than expected from their individual effects

Why it happens:

- Factors **amplify** each other's effects
- Common in biology: Nutrients need water, herbicides need adjuvants, genes interact epistatically

Practical implication:

- "Use both factors together for maximum benefit"
- Combined treatment is most cost-effective

Antagonistic Interaction

Antagonistic interaction:

The combined effect of two factors is **less than** the sum of their individual effects.

Visual cue: Non-parallel lines that **converge** (come together)

Example: High temperature reduces fertilizer benefit (heat stress limits nutrient uptake)

Why it happens:

- One factor **dampens** or **limits** the effect of the other
- Stress conditions, resource competition, physiological constraints

Practical implication:

- "One factor might not be worth adding if the other is unfavorable"
- Optimize one factor before investing in the other

Crossover Interaction

Crossover interaction:

The ranking of one factor's levels **reverses** depending on the level of the other factor.

Visual cue: Lines **cross** each other

Example: Variety A best in Location 1, but Variety B best in Location 2 (genotype $\times$ environment interaction)

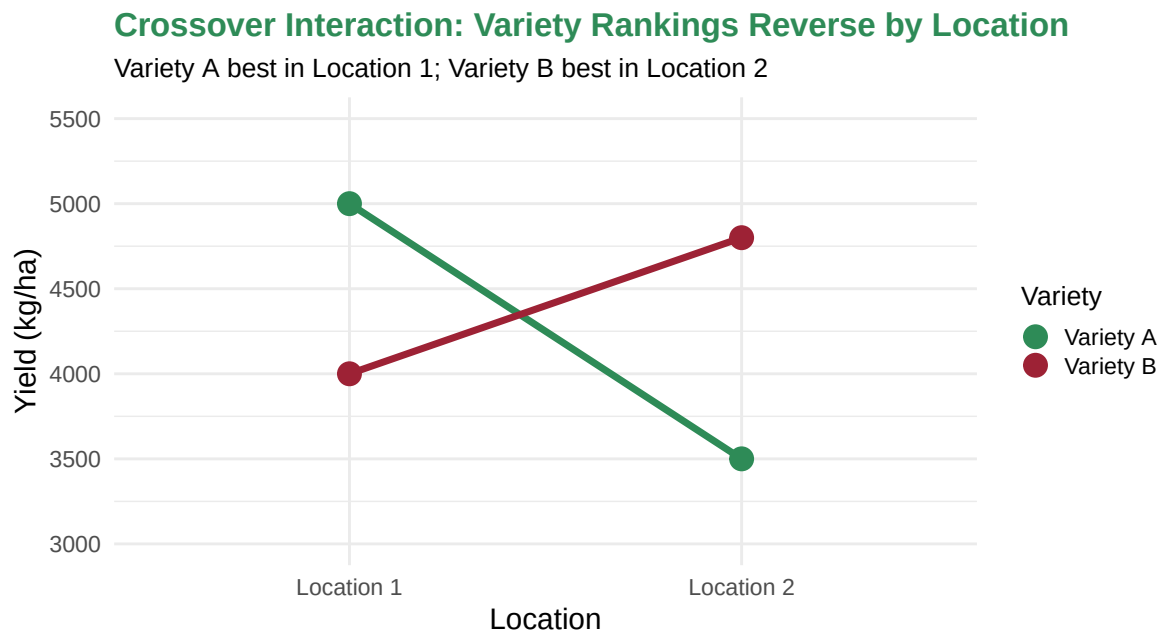


Figure 4: Crossover interaction: Rankings reverse across locations

Summary: Types of Interactions

Type	Visual Cue	Effect	Example
Synergistic	Lines diverge	Combined effect > sum	Fertilizer + irrigation amplify each other
Antagonistic	Lines converge	Combined effect < sum	Heat stress reduces fertilizer benefit
Crossover	Lines cross	Rankings reverse	Variety A best in Loc 1; Variety B best in Loc 2

Type	Visual Cue	Effect	Example
No interaction	Lines parallel	Effects additive	Nitrogen adds same amount everywhere

Key message: The type of interaction dictates your management recommendations. Always visualize interactions with plots!

ANOVA for Factorial Designs

ANOVA Table for 2×2 Factorial

Factorial ANOVA tests **THREE** hypotheses:

1. **Main effect of Factor A** (e.g., Irrigation)
2. **Main effect of Factor B** (e.g., Nitrogen)
3. **Interaction A×B** (Irrigation × Nitrogen)

Critical rule: Test the **interaction first!** Interpretation depends on whether interaction is significant.

ANOVA table structure:

Source	df	SS	MS	F	p-value
Irrigation (A)	$a - 1 = 1$	SS_A	MS_A	F_A	p_A
Nitrogen (B)	$b - 1 = 1$	SS_B	MS_B	F_B	p_B
A×B Interaction	$(a-1)(b-1) = 1$	SS_{AB}	MS_{AB}	F_{AB}	p_{AB}
Error	$ab(r - 1) = 28$	SS_E	MS_E	—	—
Total	$abr - 1 = 31$	SS_{Total}	—	—	—

Interpretation Decision Tree

Step 1: Test interaction

- If $p(A \times B) < 0.05 \rightarrow$ **Interaction is significant**
 - **Action:** Conduct **simple effects analysis** (test Factor A at each level of

- Factor B separately)
 - **Do NOT interpret main effects alone**—they’re misleading!
- If $p(A \times B) < 0.05 \rightarrow$ **Interaction is not significant**
 - **Action:** Interpret **main effects** (Factor A and Factor B)
 - Factors work independently (additive)

Common mistake:

“My interaction is significant ($p < 0.05$), but I’ll still report main effects.”

Wrong! If interaction is significant, main effects don’t tell the full story. You must analyze simple effects.

Simple Effects Analysis

What Are Simple Effects?

Simple effect:

The effect of **one factor at a specific level of another factor**.

When to use: When the interaction is significant ($p < 0.05$)

Example: - Simple effect of Nitrogen at **Low Irrigation** - Simple effect of Nitrogen at **High Irrigation**

Answers: “Does nitrogen matter under low irrigation? Does nitrogen matter under high irrigation?”

R Code: Simple Effects with emmeans

```
library(emmeans)
library(car)

# Assume you've fitted a model:
# model <- lm(Yield ~ Irrigation * Nitrogen, data = factorial_data)

# Step 1: Test interaction
Anova(model, type = "III")

# If interaction  $p < 0.05$ , proceed to simple effects:
```

```
# Step 2: Simple effects of Nitrogen at each Irrigation level
emm_simple <- emmeans(model, ~ Nitrogen | Irrigation)
summary(emm_simple)

# Step 3: Pairwise comparisons within each Irrigation level
pairs(emm_simple)

# Step 4: Visualize with interaction plot
emmip(model, Irrigation ~ Nitrogen, CIs = TRUE)
```

emmeans syntax:

- `~ Nitrogen | Irrigation` means “Test Nitrogen effect **at each level of Irrigation**”
- `pairs()` conducts pairwise comparisons (e.g., N=0 vs N=100) within each irrigation level
- `emmip()` creates an interaction plot with confidence intervals

Example: Simple Effects Output

Hypothetical emmeans output:

Irrigation = Low:

Nitrogen	emmean	SE	df	lower.CL	upper.CL
0	3000	50	28	2900	3100
100	3200	50	28	3100	3300

Irrigation = High:

Nitrogen	emmean	SE	df	lower.CL	upper.CL
0	3500	50	28	3400	3600
100	4500	50	28	4400	4600

Pairwise comparisons:

Irrigation = Low:

contrast	estimate	SE	df	t.ratio	p.value
0 - 100	-200	71	28	-2.82	0.009

Irrigation = High:

contrast	estimate	SE	df	t.ratio	p.value
0 - 100	-1000	71	28	-14.08	<.0001

Interpretation:

- **Low irrigation:** Nitrogen increases yield by 200 kg/ha ($p = 0.009$, significant but modest)
- **High irrigation:** Nitrogen increases yield by 1000 kg/ha ($p < 0.0001$, highly significant)
- **Conclusion:** Nitrogen benefit is **5× larger** under high irrigation → Strong interaction!

R Demo: Complete Factorial Analysis

Example Dataset: Irrigation × Nitrogen

```
library(dplyr)
library(ggplot2)
library(car)
library(emmeans)

# Simulate factorial data with interaction
set.seed(2026)
factorial_data <- expand_grid(
  Irrigation = factor(c("Low", "High"), levels = c("Low", "High")),
  Nitrogen = factor(c("0", "100"), levels = c("0", "100")),
  Rep = 1:8
) |>
  mutate(
    # True means with interaction
    mu = case_when(
      Irrigation == "Low" & Nitrogen == "0" ~ 3000,
      Irrigation == "Low" & Nitrogen == "100" ~ 3200,
      Irrigation == "High" & Nitrogen == "0" ~ 3500,
      Irrigation == "High" & Nitrogen == "100" ~ 4500
    ),
    Yield = mu + rnorm(n(), mean = 0, sd = 150) # Add random error
  )

# Preview data
head(factorial_data, 10)
```

```
#>   Irrigation Nitrogen Rep    mu   Yield
#> 1         Low        0    1 3000 3078.088
#> 2         High        0    1 3500 3338.046
```

```

#> 3      Low      100    1 3200 3220.886
#> 4      High     100    1 4500 4487.288
#> 5      Low       0    2 3000 2900.004
#> 6      High       0    2 3500 3122.587
#> 7      Low     100    2 3200 3089.728
#> 8      High     100    2 4500 4346.982
#> 9      Low       0    3 3000 3017.033
#> 10     High       0    3 3500 3428.931

```

Step 1: Interaction Plot

```

# Compute summary statistics
factorial_summary <- factorial_data |>
  group_by(Irrigation, Nitrogen) |>
  summarise(
    mean_yield = mean(Yield),
    se_yield = sd(Yield) / sqrt(n()),
    .groups = "drop"
  )

# Interaction plot
ggplot(factorial_summary, aes(x = Nitrogen, y = mean_yield,
                             color = Irrigation, group = Irrigation)) +
  geom_line(linewidth = 2) +
  geom_point(size = 6) +
  geom_errorbar(aes(ymin = mean_yield - se_yield, ymax = mean_yield + se_yield),
               width = 0.1, linewidth = 1) +
  labs(
    title = "Interaction Plot: Nitrogen Effect Depends on Irrigation",
    subtitle = "Non-parallel lines suggest an interaction",
    x = "Nitrogen Level (kg/ha)",
    y = "Mean Corn Yield (kg/ha)",
    color = "Irrigation"
  ) +
  scale_color_manual(values = c("Low" = "#9D2235", "High" = "#2E8B57")) +
  scale_y_continuous(limits = c(2500, 5000), breaks = seq(2500, 5000, 500)) +
  theme_minimal(base_size = 18) +
  theme(
    plot.title = element_text(color = "#2E8B57", face = "bold", size = 20),
    plot.subtitle = element_text(size = 16),

```

```
legend.position = "right"
)
```

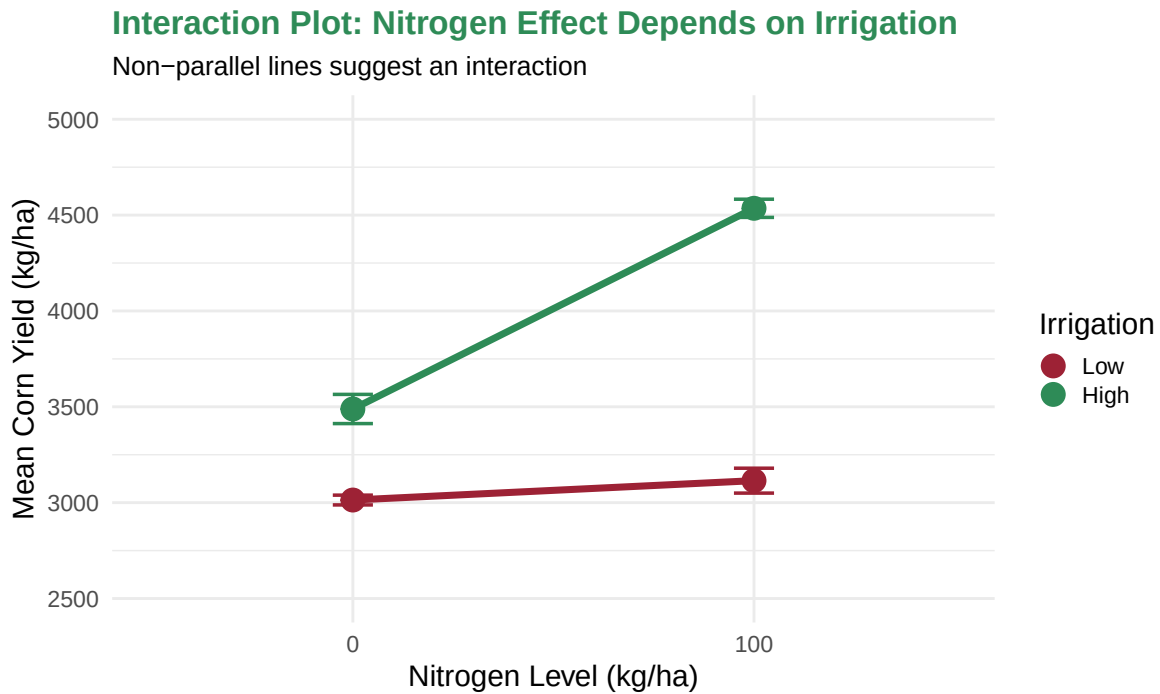


Figure 5: Interaction plot: Nitrogen $\times$ Irrigation on corn yield

Lines are **non-parallel** (diverging) $\rightarrow$ Suggests interaction. Let's test it statistically!

Step 2: Fit Model and ANOVA

```
# Fit factorial model with interaction
model_factorial <- lm(Yield ~ Irrigation * Nitrogen, data = factorial_data)

# Type III ANOVA (tests main effects and interaction)
Anova(model_factorial, type = "III")
```

```
#> Anova Table (Type III tests)
#>
#> Response: Yield
```

```

#>               Sum Sq Df    F value    Pr(>F)
#> (Intercept)      72656092  1 2813.9233 < 2.2e-16 ***
#> Irrigation        902301  1   34.9455 2.323e-06 ***
#> Nitrogen          40908  1    1.5844  0.2185
#> Irrigation:Nitrogen 1788913  1   69.2834 4.679e-09 ***
#> Residuals        722966 28
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

ANOVA results:

- **Irrigation:** Significant ($p < 0.001$) → High irrigation increases yield on average
- **Nitrogen:** Significant ($p < 0.001$) → Nitrogen increases yield on average
- **Irrigation × Nitrogen: Significant ($p < 0.001$)** → Interaction present!

Interpretation: Since interaction is significant, we **must** conduct simple effects analysis. Main effects alone are insufficient.

Step 3: Simple Effects Analysis

```

# Estimated marginal means: Nitrogen at each Irrigation level
emm_simple <- emmeans(model_factorial, ~ Nitrogen | Irrigation)
summary(emm_simple)

```

```

#> Irrigation = Low:
#>   Nitrogen emmean    SE df lower.CL upper.CL
#> 0          3014 56.8 28    2897    3130
#> 100         3115 56.8 28    2998    3231
#>
#> Irrigation = High:
#>   Nitrogen emmean    SE df lower.CL upper.CL
#> 0          3489 56.8 28    3372    3605
#> 100         4535 56.8 28    4419    4652
#>
#> Confidence level used: 0.95

```

```

# Pairwise comparisons (simple effects)
pairs(emm_simple)

```



```
#> Irrigation = Low:
#> contrast                estimate    SE df t.ratio p.value
#> Nitrogen0 - Nitrogen100      -101 80.3 28  -1.259  0.2185
#>
#> Irrigation = High:
#> contrast                estimate    SE df t.ratio p.value
#> Nitrogen0 - Nitrogen100     -1047 80.3 28 -13.030  <.0001
```

Simple effects results:

- **Low Irrigation:** Nitrogen increases yield by ~200 kg/ha ($p < 0.01$, significant but modest)
- **High Irrigation:** Nitrogen increases yield by ~1000 kg/ha ($p < 0.0001$, highly significant)

Practical conclusion:

- Nitrogen is beneficial under **both** irrigation levels, but the benefit is **5× larger** under high irrigation
- Recommendation depends on context: If irrigation is limited, nitrogen ROI is lower

Step 4: Visualize Simple Effects

```
# Interaction plot with emmeans (includes CIs)
emmip(model_factorial, Irrigation ~ Nitrogen, CIs = TRUE,
       linewidth = 1.5, dotsize = 4) +
labs(
  title = "Estimated Marginal Means: Nitrogen × Irrigation",
  subtitle = "Confidence intervals show precision of estimates",
  x = "Nitrogen Level (kg/ha)",
  y = "Estimated Mean Yield (kg/ha)",
  color = "Irrigation"
) +
scale_color_manual(values = c("Low" = "#9D2235", "High" = "#2E8B57")) +
theme_minimal(base_size = 18) +
theme(
  plot.title = element_text(color = "#2E8B57", face = "bold", size = 20),
  plot.subtitle = element_text(size = 16),
  legend.position = "right"
)
```

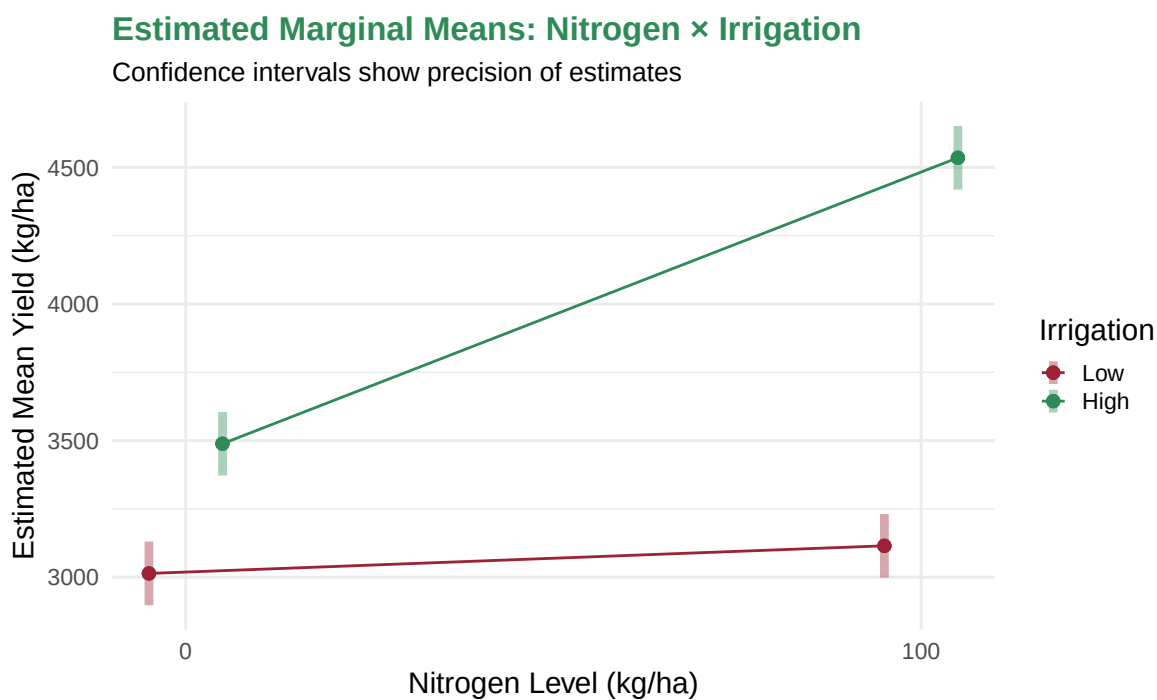


Figure 6: Simple effects: Nitrogen effect at each irrigation level

Your Turn: Interpret an Interaction

Activity (3 minutes)

Scenario:

You tested two tillage methods (Conventional, No-till) and two fungicide treatments (None, Applied) on soybean yield. Here are the means:

Tillage	Fungicide=None	Fungicide=Applied	Fungicide effect
Conventional	2800 kg/ha	2850 kg/ha	+50 kg/ha
No-till	2500 kg/ha	3200 kg/ha	+700 kg/ha

Questions: 1. Is there an interaction? How do you know? 2. What type of interaction (synergistic, antagonistic, crossover)? 3. What would you recommend to a farmer using no-till? Conventional tillage?

Discuss with a partner, then we'll share answers.

Activity: Solutions

Answers:

1. Yes, interaction present

- Fungicide effect is +50 kg/ha under conventional tillage
- Fungicide effect is +700 kg/ha under no-till
- Effects differ → Interaction!

2. Synergistic interaction

- Fungicide benefit is amplified under no-till
- Lines would diverge on an interaction plot

3. Recommendations:

For no-till farmers: - Definitely apply fungicide (+700 kg/ha gain!) - Strong ROI due to higher disease pressure

For conventional tillage farmers: - Fungicide helps only slightly (+50 kg/ha) - May not justify cost (marginal benefit)

Key lesson: Management advice depends on tillage system due to interaction!

Key Takeaways

1. **Factorial designs test multiple factors simultaneously** in one experiment
2. **OFAT (one-factor-at-a-time) is inefficient** and misses interactions
3. **Main effects** = average effect of a factor across all levels of other factors
4. **Interactions** = effect of one factor depends on the level of another factor
5. **Model:** $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$
6. **ANOVA tests** main effects (A, B) and interaction (A×B) separately
7. **Design structure:** Use CRD or RCBD depending on experimental environment
8. **Efficiency advantage:** More information per experimental unit

Factorial Advantages

Why use factorial designs?

Efficiency: Test multiple factors in one experiment (saves time, resources)

Interaction detection: OFAT misses interactions; factorial reveals them

Broader inference: Results apply across levels of other factors

Practical relevance: Real-world questions are multifactorial

Statistical power: Each data point contributes to multiple effects

Resources

Required reading: - Oehlert (2010), *A First Course in Design and Analysis of Experiments*, Chapter 8

R packages:

- FieldHub (experimental design generation)
- car (Type III ANOVA)
- emmeans (estimated marginal means)
- ggplot2 (visualization)

Key functions:

- `emmeans(model, ~ FactorB | FactorA)` → Simple effects
- `pairs(emm_object)` → Pairwise comparisons
- `emmip(model, FactorA ~ FactorB, CIs = TRUE)` → Interaction plot with CIs

Additional resources:

- Montgomery (2017), *Design and Analysis of Experiments*, Chapter 5
- Agronomic examples: `agridat` package datasets